

Southwest Airlines, a staple of the southwest airline industry, like all airlines use a variety of metrics to assess the status of the company. A key metric used by the airline industry are revenue passenger miles (RPM) which measure the demand for air transportation; it equals the number of revenue-paying passengers multiplied by the distance they travel. Forecasting this metric is an essential task for every airline analyst.

In what follows, I consider a monthly model of the natural log of total revenue passenger miles.

I. Preliminary Analysis

Total revenue passenger miles (r_t) are given at a monthly value between 2002 Jan to 2025 Jan. The data used in this analysis is the series labeled “Southwest” taken from the “Airline Total RPM.xlsx” dataset.

Panel (a) of Figure 1 shows the total revenue passenger miles from 2002 January to 2025 January.

- The series shows an indisputable upwards trend from the beginning to end of sample.
- Volatility in the trend suggests a seasonal component to the data. There also appears to be a change in volatility around 2020 January.

→ As such, it is reasonable to assume that the r_t series is not covariance stationary.

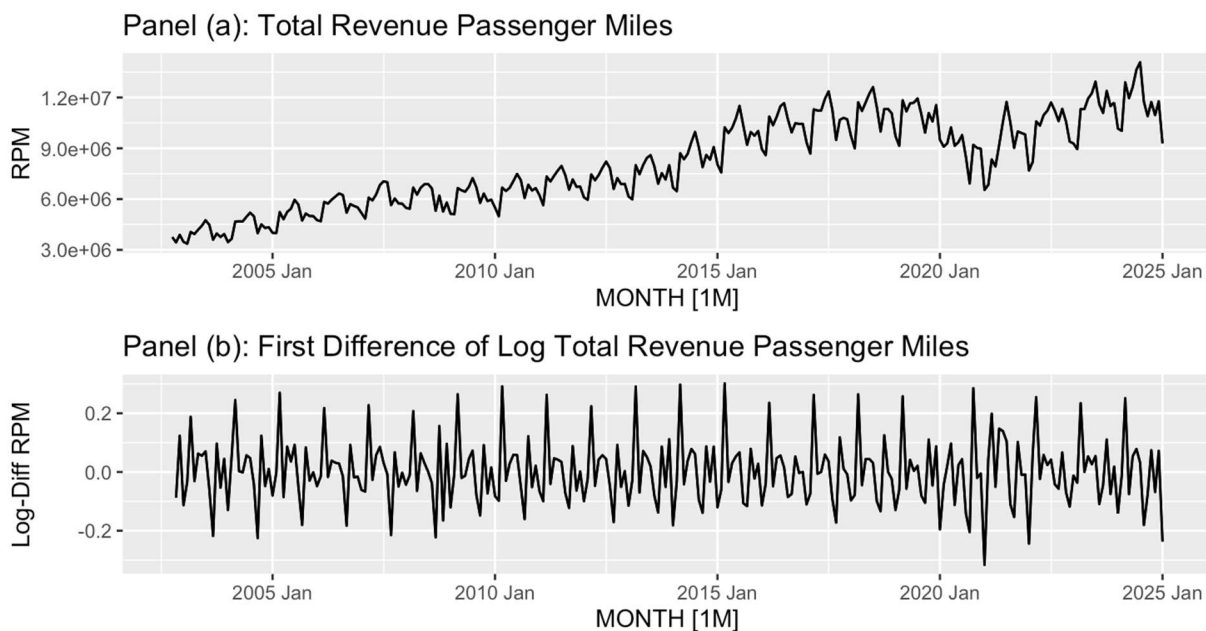


Figure 1. Time Path of Total Revenue Passenger Miles

Taking an appropriate measure to achieve stationarity, I log the r_t to dampen the series and take the first difference. Panel (b) in Figure 1 shows that by doing so the new Δr_t series appears to be stationary at first glance. The seasonal components pointed out in Panel (a) are no longer as apparent. Upon closer inspection of the Δr_t series, I suspect seasonality persists beyond the first difference. To verify my suspicion, I run a HEGY test on the Δr_t series with only a constant and am unable to reject the presence of seasonal roots between months.

Considering the possibility of seasonal roots, I take the Δr_t series and visualize 23 years of data for each month in Figure 2. Notably, the average of each month is different and thus non-stationary for the first difference of log RPM. At this point, I consider two approaches to achieving stationarity:

- Take a non-seasonal and seasonal difference, essentially double differencing with the hope the data is not reduced to non-forecastable data.
- Seasonally difference the series to get rid of seasonality and possibly trend.

I pursue the second option, seasonally differencing, but verifying with an ADF and HEGY tests tell me that I cannot fully reject the presence of a trend unit root.

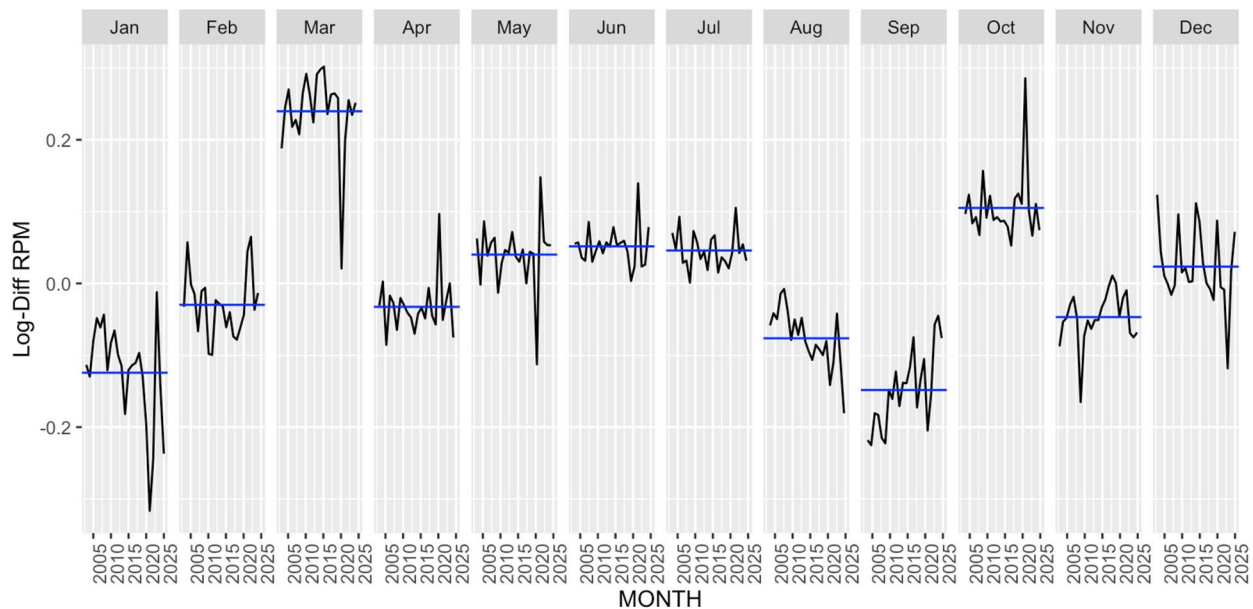


Figure 2. Monthly Seasonality of First Difference of Log RPM

The first option of taking a non-seasonal and seasonal difference is what ultimately achieves stationarity, as seen in Figure 3. Verifying the claim, I run ADF, KPSS, and HEGY tests which all validate my assumptions of stationarity and no unit roots. The double-differenced $\Delta_1 \Delta_{12} r_t$ series now appears to be white noise with volatility spikes clustered together. It is now reasonable to estimate a model of the $\Delta_1 \Delta_{12} r_t$ series without any further transformations.

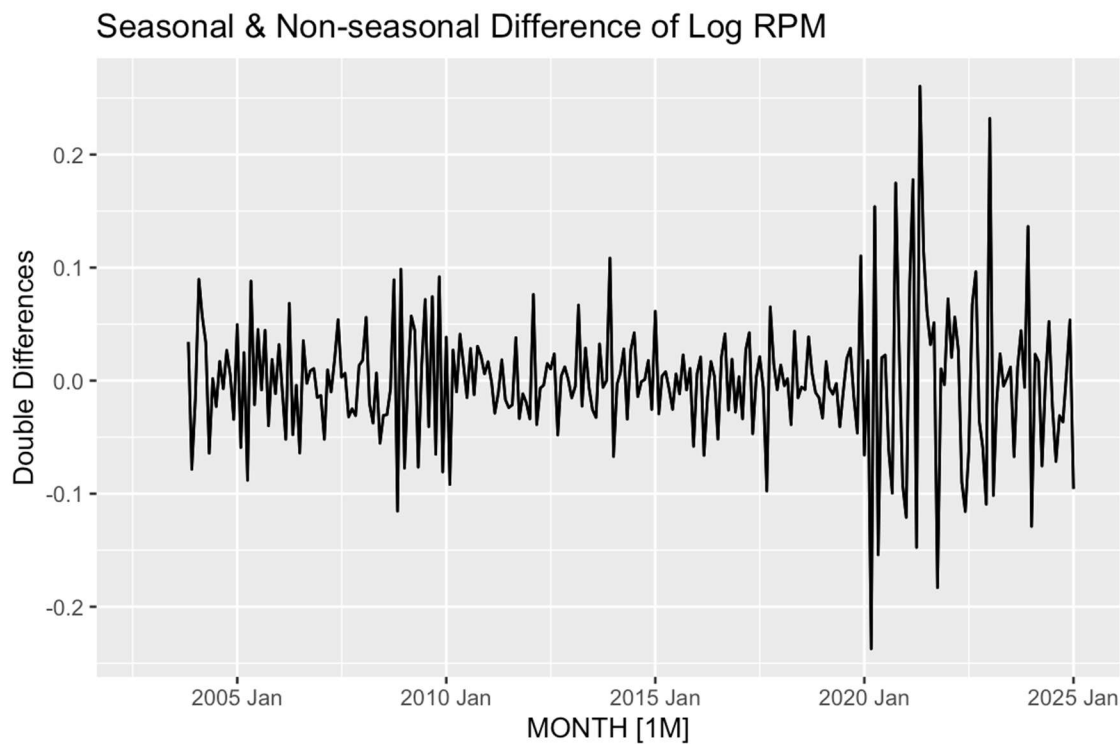


Figure 3. Double Differenced Log of RPM

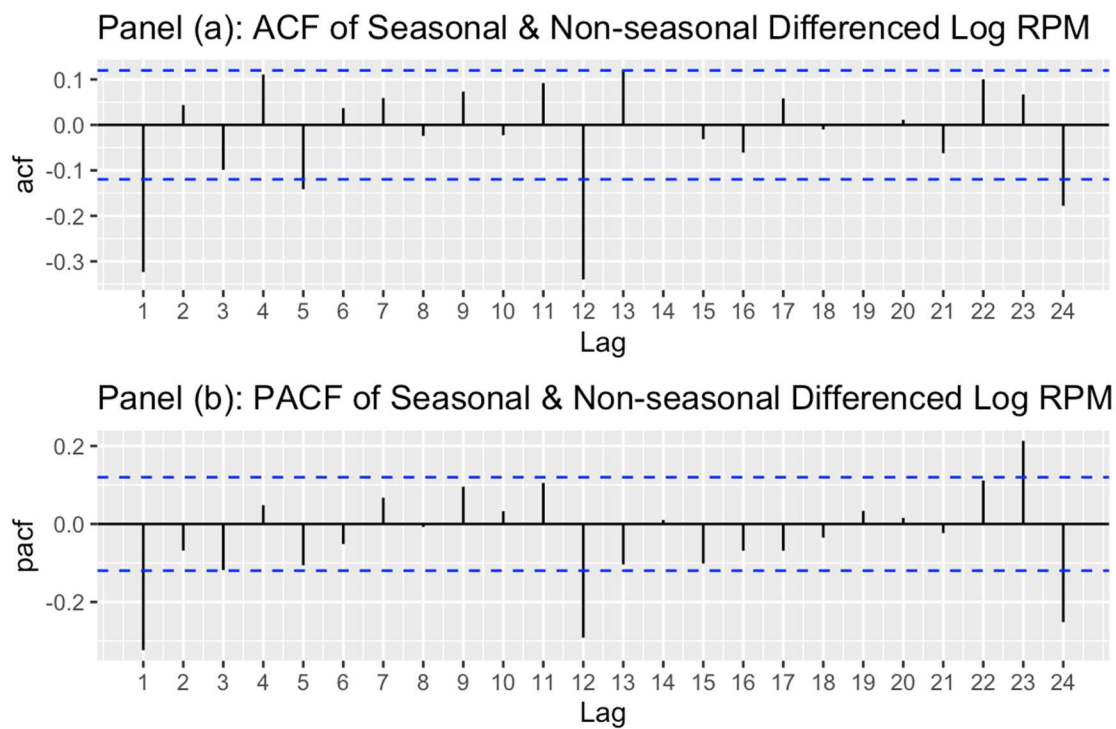


Figure 4. (P)ACF of the Total Revenue Passenger Miles

I now examine the autocorrelation and partial autocorrelation functions of the $\Delta_1 \Delta_{12} r_t$ series in Figure 4. The (P)ACF of the untransformed r_t series is in the appendix.

- The correlogram confirms stationarity of the r_t series: ACF and PACF quickly decay.
- Both ACF and PACF have recurring significant lags at yearly intervals

As both correlograms do not geometrically decay, AR(p) and MA(q) processes could be present in the model. Examining Panel (a) for MA(q) process, there is an immediate cut off after lag 1 and recurring significant lags at 12 and 24. This suggests at least a seasonal MA component, possibly a MA as well. The largest significant lags are at the beginning, approximately .30, and gradually decay.

Panel (b) the PACF, like the ACF, cuts off after one lag and does not geometrically decay. Lags at 12 intervals are significant and could be interpreted as seasonal AR(p) components. A significant lag at 23 could indicate a restricted AR(23) component. I forgo the idea as it is small, approximately .20, and significantly far from the first lag. The largest significant lags are at the beginning, approximately .30, and gradually decay.

In comparison, the untransformed r_t series (P)ACF in Figure A1 Panel (a) shows gradual decay over a long period of time. Panel (b) shows an immediate cut off from lag 1, but oscillating features every 6 lags, possibly seasonally affected lags. Both panels conform to the theoretical characteristics of a stationary time series.

As only (S)MA(1) and (S)AR(1) components could be identified as significant, I will attempt multiple alternative ARMA specifications.

II. Estimation & Diagnostics

Given the r_t series required to be double-differenced, seasonally and non-seasonally, I start by assuming a SARIMA model. Estimating a total of five different SARIMA process, Table 1 outlines all estimate at their respective columns, the t-statistics, information criteria, and Q-statistic for $K = 6, 12, 24$. Given the information from the (P)ACF in Figure 4 I conclude the usage of at most 2 (S)MA and (S)AR lags as an attempt to account for any misspecification.

Estimating the five SARIMA models below shows that the immediate standout model is SARIMA(0,1,1)(1,1,1) has the second lowest information criteria in all categories. The model also has significant t-statistics at the 5 percent level (although not that important when using both AR and MA

components). The closest model in terms of composition is SARIMA(0,1,1)(2,1,1) which shares the same strengths as the first model. The Q-statistics at all tested lags indicate that the residuals exhibit no autocorrelation at any conventional significance level in either model..

Assuming the maximum usage of (S)AR and (S)MA components, the SARIMA(2,1,2)(2,1,2) model performs worse than the less complex models, possibly due to overfitting. Only the AR_2 and MA_2 coefficients are statistically significant. Notably the AR_1 and AR_2 coefficients almost cancel each other out. In other model specification I exclude AR components for this reason and relegate the evidence of the PACF to (S)AR lags along with reducing the number of MA lags to 1. The new estimated model is SARIMA(0,1,1)(2,1,2) and performs marginally better than the previous model in information criteria. The Q-statistics indicate at all tested lags there is no autocorrelation like the previous model. Both models are eliminated from consideration.

Table 1. SARIMA Model Estimates of Log RPM

	SARIMA (2,1,2) (2,1,2)	SARIMA(0,1,1) (1,1,1)	SARIMA(0,1,1) (2,1,1)	SARIMA(2,1,1) (2,1,2)	SARIMA(0,1,1) (2,1,2)
AR_1	-0.3115 (-1.01)			0.3596 (1.65)	
AR_2	0.3419 (2.14)			0.1758 (1.50)	
MA_1	-0.0495 (-0.163)	-0.3615 (-5.56)	-0.3368 (-5.75)	-0.7380 (-3.62)	-0.03664 (-5.74)
MA_2	-0.4007 (-2.80)				
SAR_1	-0.4662 (-1.18)	0.1196 (1.27)	-0.0040 (-0.0332)	-0.3922 (-0.735)	0.1061 (0.125)
SAR_2	-0.1307 (-0.924)		-0.2098 (-1.96)	-0.1616 (-1.07)	-0.2232 (-1.55)
SMA_1	-0.2425 (-0.612)	-0.8367 (-13.5)	-0.7075 (-6.79)	-0.3182 (-0.591)	-0.8187 (-0.957)
SMA_2	-0.4000 (-1.16)			-0.3300 (-0.710)	0.0940 (-0.131)
α	-7E-04 (-0.900)	-3E-04 (-0.698)	-5E-04 (-0.823)	-3E+04 (-0.933)	-5E-04 (-0.820)
AIC	-751.11	-752.44	-754.19	-750.33	-752.21
AICc	-750.11	-752.17	-753.82	-749.52	-751.71
BIC	-716.69	-735.23	-733.54	-719.35	-728.11
Q(6) p-value	0.9509	0.2425	0.2716	0.6153	0.2795
Q(12) p-value	0.9334	0.5632	0.5790	0.8161	0.5873
Q(24) p-value	0.9820	0.7026	0.8767	0.9502	0.8833

NOTES: t-statistics in parenthesis

Choosing to approach model specification in the opposite direction I chose to keep the AR lags and removed MA_2 , this model specification is the worst among them all. There is only 1 significant lag, MA_2 and AR_1 and AR_2 do not cancel each other out like in the $SARIMA(2,1,2)(2,1,2)$ model. In regards to that model, $SARIMA(2,1,1)(2,1,2)$ is a worse estimation in almost every category except the BIC. This model is eliminated from consideration.

The “best” model estimated is $SARIMA(0,1,1)(1,1,1)$ but I do not want to rely on only one model, so $SARIMA(0,1,1)(2,1,1)$ will also be forecasted. Both models can be considered the “best” of the models forecasted.

III. Out of Sample Forecast

I assess the forecasting performance of the $SARIMA(0,1,1)(1,1,1)$ and $SARIMA(0,1,1)(2,1,1)$ models by examine their variance prediction error, and mean squared prediction error.

An obstacle in deciding the training period of the dataset was whether to include the Covid pandemic. Given the exogenous nature of the shock, it is unlikely that training the model on data pre-2020 will produce an accurate forecast when there has been no similar shock in the history of the data. I formulated two options:

- Train the data from 2002 October to 2023 January to include the Covid shock and forecast out of sample 24 months.
- Train the data from 2002 October to 2020 January and forgo any possibility of producing an accurate forecast of Covid and hope post-Covid forecasts are accurate 60 months out.

The first option does not follow the traditionally suggested 80-20 ratio, I choose approximately 10 percent of the data as my holdout. A 24-month holdout sample is, in my opinion, a good enough forecasting sample for assessing forecast accuracy. The further out I forecast the probability of being accurate diminishes.

First, I estimate the two models with a 1-step ahead forecast and an additional 23 1-step ahead forecasts are obtained without re-estimating the coefficients as seen in Figure 5. To evaluate the accuracy of the forecast I examine the first forecast where the actual value $r_{2025Feb} = 16.00706$. The $SARIMA(0,1,1)(1,1,1)$ model predicted a value of 15.99280 and $SARIMA(0,1,1)(2,1,1)$ predicted a value of 15.98343. Both models are close to the actual value and each other. Deciding which forecast is superior cannot be assessed using the first predicted value. Instead, I will look at variance prediction errors for both models.

Let e_{1t} denote the prediction error from SARIMA(0,1,1)(1,1,1) and e_{2t} from SARIMA(0,1,1)(2,1,1). The mean of e_{1t} is -0.0001 and e_{2t} is 0.0065. The $VPE(e_{1t}) = 0.0014$ and $VPE(e_{2t}) = 0.0014$. Likewise, the $MSPE(e_{1t}) = 0.0013$ and $MSPE(e_{2t}) = 0.0014$. Finally, I assess both models with the Diebold-Mariano test for predictive accuracy, $DM = -0.50587$ with a p-value = 0.6178, thus the MSPE of both models are not statically different from each other.

Both models have proven themselves worthwhile to forecast and should both be considered.

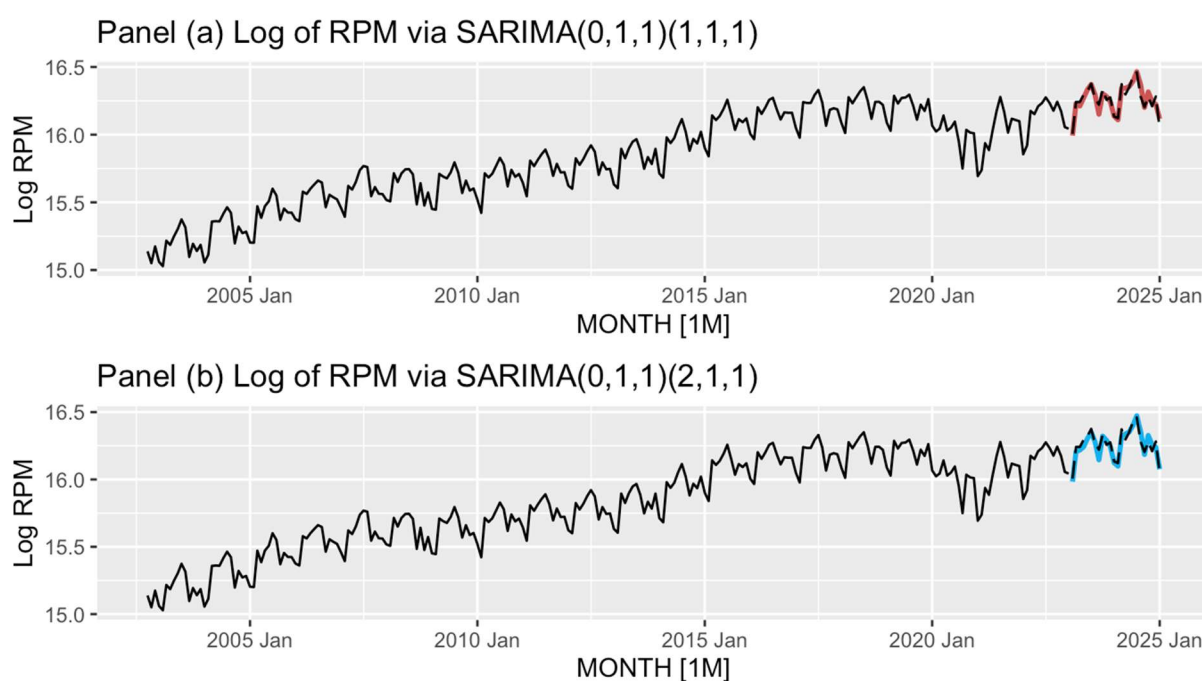


Figure 5. The 1-step ahead Forecasts of Log RPM in 2023 Feb – 2025 Jan

Disclaimer: The use of Artificial Intelligence was limited to debugging syntax errors in RStudio.

Appendix

